Evaluation runs: 2pc-ab (two-pc-abstain-kragl)

generated 2026-11-03 from brain-data b2ae60f86 2026-11-03 17:57:09 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): no entry in the rule file (the README states the expectation). Blue rows are at the current pins (TraceForge 9a0629e, merge join 367cdc6, JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". G_{spec}, G_{impl} = the graphs of the specification and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is G^{expl}_{spec}); G_x = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

stateless

stateless: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G^{expl}_{spec}	run
1	7	3	conforms	0.0	2	1 h, 250 GB	3	8	brain03, 2026-11-01
2	36	13	conforms	0.0	2	1 h, 250 GB	9	138	brain03, 2026-11-01
3	238	67	conforms	0.3	2	1 h, 250 GB	44	4838	brain03, 2026-11-01
4	1962	377	conforms	23.3	2	1 h, 250 GB	175	177332	brain03, 2026-11-01
5	19744	2291	conforms	445.5	2	1 h, 250 GB	675	7157806	brain03, 2026-11-01
6	236938	15205	conforms	229.0	10	1 h, 250 GB	739	3830608	brain03, 2026-11-01
7	3314280		🕓 (1 h)			1 h, 250 GB			brain03, 2026-11-01

stateless: pair **off-by-one**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	sweeps	G^{expl}_{spec}	run
1	7	3	2	violation	0.0	2	1 h, 250 GB	3	15	brain03, 2026-11-01
2	36	13	6	violation	0.0	2	1 h, 250 GB	12	271	brain03, 2026-11-01
3	238	67	18	violation	0.2	2	1 h, 250 GB	41	5269	brain03, 2026-11-01
4	1962	377	72	violation	16.9	2	1 h, 250 GB	241	314812	brain03, 2026-11-01
5	19744	2291	360	violation	278.7	4	1 h, 250 GB	598	7443355	brain03, 2026-11-01
6	236938			🕓 (1 h)			1 h, 250 GB			brain03, 2026-11-01

Not run (stateless). skipped: a smaller size had reached its limit: 7 off-by-one bag, 8 correct bag, 8 off-by-one bag, 9 correct bag, 9 off-by-one bag, 10 correct bag, 10 off-by-one bag, 11 off-by-one bag, 11 correct bag.

planned but never scheduled (grid points beyond the ladder): 8 correct bag, 8 off-by-one bag, 9 correct bag, 9 off-by-one bag, 10 correct bag, 10 off-by-one bag.

stateful

stateful: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
1	7	3	conforms	0.9	2	1 h, 250 GB	brain02, 2026-11-03
2	36	13	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-03
3	238	67	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-03
4	1962	377	conforms	0.2	2	1 h, 250 GB	brain02, 2026-11-03
5	19744	2291	conforms	1.5	4	1 h, 250 GB	brain02, 2026-11-03
6	236938	15205	conforms	8.0	44	1 h, 250 GB	brain02, 2026-11-03

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
7	3314280	111539	conforms	63.3	238	1 h, 250 GB	brain02, 2026-11-03
8	52994722	909809	conforms	491.9	872	48 h, 250 GB	brain05, 2026-11-02
9	953561800	8247331	conforms	11686.0	3022	48 h, 250 GB	brain05, 2026-11-02

stateful: pair **off-by-one**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	run
1	7	3	2	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-03
2	36	13	6	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-03
3	238	67	18	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-03
4	1962	377	72	violation	0.2	2	1 h, 250 GB	brain02, 2026-11-03
5	19744	2291	360	violation	1.2	8	1 h, 250 GB	brain02, 2026-11-03
6	236938	15205	2160	violation	7.1	42	1 h, 250 GB	brain02, 2026-11-03
7	3314280	111539	15120	violation	55.7	242	1 h, 250 GB	brain02, 2026-11-03
8	52994722	909809	120960	violation	532.8	853	48 h, 250 GB	brain05, 2026-11-02
9	953561800	8247331	1088640	violation	11598.0	3024	48 h, 250 GB	brain05, 2026-11-02

Not run (stateful). left out by the table rules (ok): 8 correct bag, 8 off-by-one bag.
left out by the table rules (timeout): 9 correct bag, 9 off-by-one bag.
skipped: a smaller size had reached its limit: 10 off-by-one bag, 10 correct bag, 11 off-by-one bag, 11 correct bag.
planned but never scheduled (grid points beyond the ladder): 10 correct bag, 10 off-by-one bag, 11 off-by-one bag, 11 correct bag, 12 correct bag, 12 off-by-one bag.

offline

offline: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	7	3	conforms	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
2	36	13	conforms	0.1	4	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain05, 2026-11-02
3	238	67	conforms	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
4	1962	377	conforms	0.3	6	10 min, 16 GB	0.2	0.1	0.0	in-memory	brain05, 2026-11-02
5	19744	2291	conforms	1.4	22	10 min, 16 GB	1.0	0.2	0.1	in-memory	brain05, 2026-11-02
6	236938	15205	conforms	7.0	138	10 min, 16 GB	5.2	0.8	0.9	in-memory	brain05, 2026-11-02
7	3314280	111539	conforms	51.4	161	1 h, 250 GB	42.9	3.4	5.1	merge	brain05, 2026-11-03
8	52994722	909809	conforms	612.2	787	48 h, 250 GB	576.0	13.3	23.0	merge	brain03, 2026-11-03
9	953561800	8247331	conforms	11474.2	4704	48 h, 250 GB	11234.0	109.3	130.9	merge	brain03, 2026-11-03

offline: pair **off-by-one**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	7	3	2	violation	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
2	36	13	6	violation	0.1	4	10 min, 16 GB	0.0	0.0	0.0	in-memory	brain05, 2026-11-02
3	238	67	9	violation	0.2	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
4	1962	377	12	violation	0.3	6	10 min, 16 GB	0.2	0.1	0.0	in-memory	brain05, 2026-11-02
5	19744	2291	15	violation	1.4	22	10 min, 16 GB	1.0	0.2	0.2	in-memory	brain05, 2026-11-02
6	236938	15205	18	violation	6.9	36	1 h, 250 GB	5.2	0.9	0.8	merge	brain05, 2026-11-03
7	3314280	111539	21	violation	51.8	759	10 min, 16 GB	42.9	3.3	5.5	in-memory	brain05, 2026-11-02
8	52994722	909809	24	violation	611.8	787	48 h, 250 GB	576.0	11.8	24.0	merge	brain03, 2026-11-03
9	953561800	8247331	27	violation	11487.6	4704	48 h, 250 GB	11234.0	121.4	132.2	merge	brain03, 2026-11-03

Not run (offline). planned but never scheduled (grid points beyond the ladder): 8 correct bag, 8 off-by-one bag, 9 correct bag, 9 off-by-one bag, 10 correct bag, 10 off-by-one bag, 11 off-by-one bag, 11 correct bag.
refused (an input missing at run time): 10 off-by-one bag, 10 correct bag.
skipped: a smaller size had reached its limit: 11 off-by-one bag, 11 correct bag.

cross offline, Java implementation against the Rust specification

cross offline, Java implementation against the Rust specification: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	7	3	conforms	3.1	172	10 min, 16 GB	0.1	3.1	0.0	in-memory	brain01, 2026-11-02
2	36	13	conforms	3.2	156	10 min, 16 GB	0.0	3.1	0.0	in-memory	brain01, 2026-11-02
3	238	67	conforms	3.6	170	10 min, 16 GB	0.1	3.5	0.0	in-memory	brain01, 2026-11-02

cross offline, Java implementation against the Rust specification: pair **off-by-one**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	7	3	2	violation	3.2	160	10 min, 16 GB	0.1	3.1	0.1	in-memory	brain01, 2026-11-02
2	36	13	6	violation	3.4	166	10 min, 16 GB	0.0	3.4	0.0	in-memory	brain01, 2026-11-02
3	238	67	9	violation	3.5	158	10 min, 16 GB	0.1	3.4	0.0	in-memory	brain01, 2026-11-02

Not run (cross offline, Java implementation against the Rust specification). refused (an input missing at run time): 4 off-by-one bag, 4 correct bag, 5 correct bag, 5 off-by-one bag.

planned but never scheduled (grid points beyond the ladder): 8 correct bag, 8 off-by-one bag, 9 correct bag, 9 off-by-one bag, 10 correct bag, 10 off-by-one bag.

cross offline, Rust implementation against the Java specification

cross offline, Rust implementation against the Java specification: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	7	3	conforms	3.2	156	10 min, 16 GB	3.1	0.1	0.0	in-memory	brain01, 2026-11-02
2	36	13	conforms	3.4	156	10 min, 16 GB	3.4	0.1	0.0	in-memory	brain01, 2026-11-02
3	238	67	conforms	4.0	170	10 min, 16 GB	4.0	0.1	0.0	in-memory	brain01, 2026-11-02
4	1962	377	conforms	9.4	753	10 min, 16 GB	9.2	0.1	0.0	in-memory	brain01, 2026-11-02
5	19744	2291	conforms	73.1	1733	10 min, 16 GB	72.7	0.2	0.1	in-memory	brain01, 2026-11-02

cross offline, Rust implementation against the Java specification: pair **off-by-one**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	7	3	2	violation	3.2	156	10 min, 16 GB	3.1	0.1	0.0	in-memory	brain01, 2026-11-02
2	36	13	6	violation	3.4	156	10 min, 16 GB	3.4	0.0	0.0	in-memory	brain01, 2026-11-02
3	238	67	9	violation	4.1	170	10 min, 16 GB	4.0	0.1	0.0	in-memory	brain01, 2026-11-02
4	1962	377	12	violation	9.3	753	10 min, 16 GB	9.2	0.1	0.0	in-memory	brain01, 2026-11-02

Not run (cross offline, Rust implementation against the Java specification). skipped: a smaller size had reached its limit: 5 off-by-one bag.

planned but never scheduled (grid points beyond the ladder): 8 correct bag, 8 off-by-one bag, 9 correct bag, 9 off-by-one bag, 10 correct bag, 10 off-by-one bag.

cross online, Java implementation (JMC) against the Rust specification export

cross online, Java implementation (JMC) against the Rust specification export: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	7	3	conforms	0.6	146	1 h, 250 GB		brain05, 2026-11-02
2	36	13	conforms	0.7	175	1 h, 250 GB		brain05, 2026-11-02
3	238	67	conforms	0.9	166	1 h, 250 GB		brain05, 2026-11-02
4	1962	377	conforms	2.0	287	1 h, 250 GB		brain05, 2026-11-02
5	19744	2291	conforms	9.2	864	1 h, 250 GB		brain05, 2026-11-02

cross online, Java implementation (JMC) against the Rust specification export: pair off-by-one, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
1	7	3	2	violation	0.6	151	1 h, 250 GB	74	brain05, 2026-11-02
2	36	13	6	violation	0.7	146	1 h, 250 GB	42	brain05, 2026-11-02
3	238	67	18	violation	1.0	163	1 h, 250 GB	58	brain05, 2026-11-02
4	1962	377	72	violation	1.9	275	1 h, 250 GB	50	brain05, 2026-11-02
5	19744	2291	360	violation	9.7	947	1 h, 250 GB	54	brain05, 2026-11-02

Not run (cross online, Java implementation (JMC) against the Rust specification export). planned but never scheduled (grid points beyond the ladder): 1 correct bag, 1 off-by-one bag, 2 off-by-one bag, 2 correct bag, 3 off-by-one bag, 3 correct bag, 4 correct bag, 4 off-by-one bag, 5 off-by-one bag, 5 correct bag, 8 correct bag, 8 off-by-one bag, 9 correct bag, 9 off-by-one bag, 10 correct bag, 10 off-by-one bag.

cross online, Rust implementation (TraceForge) against the Java specification export

cross online, Rust implementation (TraceForge) against the Java specification export: pair correct, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	7	3	conforms	0.0	4	1 h, 250 GB		brain04, 2026-11-01
2	36	13	conforms	0.0	4	1 h, 250 GB		brain04, 2026-11-01
3	238	67	conforms	0.0	4	1 h, 250 GB		brain04, 2026-11-01
4	1962	377	conforms	0.1	6	1 h, 250 GB		brain04, 2026-11-01
5	19744	2291	conforms	0.4	28	1 h, 250 GB		brain04, 2026-11-01

cross online, Rust implementation (TraceForge) against the Java specification export: pair off-by-one, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
1	7	3	2	violation	0.0	4	1 h, 250 GB	1	brain04, 2026-11-01
2	36	13	6	violation	0.0	4	1 h, 250 GB	2	brain04, 2026-11-01
3	238	67	18	violation	0.0	4	1 h, 250 GB	3	brain04, 2026-11-01
4	1962	377	72	violation	0.1	8	1 h, 250 GB	3	brain04, 2026-11-01
5	19744	2291	360	violation	0.4	26	1 h, 250 GB	27	brain04, 2026-11-01

Not run (cross online, Rust implementation (TraceForge) against the Java specification export). left out by the table rules (ok): 1 correct bag, 1 off-by-one bag, 2 off-by-one bag, 2 correct bag, 3 off-by-one bag, 3 correct bag, 4 correct bag, 4 off-by-one bag, 5 off-by-one bag, 5 correct bag.

planned but never scheduled (grid points beyond the ladder): 8 correct bag, 8 off-by-one bag, 9 correct bag, 9 off-by-one bag, 10 correct bag, 10 off-by-one bag.